(10 Marks)

Q: Explain the A* search algorithm in detail with a neat example. Discuss its working, heuristic function, admissibility and advantages and limitations.

Ans:

1) **Introduction:**
   - A* (A-star) is an informed search algorithm.
   - Used to find the shortest (least cost) path from a start node to a goal node.
   - It uses both actual cost and a heuristic estimate to guide the search.

2) **Evaluation function:**

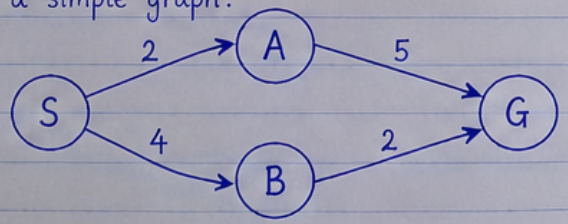
$$f(n) = g(n) + h(n)$$

   - g(n) : cost from start node to node n.
   - h(n) : estimated cost from node n to goal node.
   - f(n) : estimated total cost.

3) **Working (Steps):**
   - Keep an OPEN list (frontier) and CLOSED list.
   - Select the node with the smallest f(n).
   - If the node is goal node, stop and return the path.
   - Otherwise, expand the node and generate its successors.
   - Update g(n), h(n) and f(n) for successors.
   - Repeat until goal is reached.

4) **Example:**

Consider a simple graph:



Using A*, the algorithm will expand nodes based on f(n) and find the shortest path from S to G.

5) **Heuristic function:**
   - h(n) gives an estimate of the cost from a node to the goal node.
   - It helps in choosing the most promising node.
   - Different heuristics can be used depending on the problem (e.g., distance, Manhattan distance etc.).

6) **Admissibility:**
   - A heuristic is admissible if it never overestimates the actual cost to reach the goal.
   - If the heuristic is admissible, A* is guaranteed to find the optimal path.

7) **Advantages:**
   - Finds optimal solution (with admissible heuristic).
   - More efficient than uninformed search algorithms (e.g., BFS, DFS).
   - Widely used in navigation, robotics and game playing.

8) **Limitations:**
   - Requires more memory to store nodes.
   - Performance depends on the quality of the heuristic function.
   - Can be slow for very large search spaces.

9) **Conclusion:**
   - A* is a useful algorithm that combines actual cost and heuristic estimate to find an efficient path from a start node to a goal node.